

BTP Progress Report-1

Topological Analysis of Plant Structure Using Persistent Homology

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1. Abstract

This semester, my focus was on understanding, implementing, and experimentally validating a topological data analysis (TDA)-based framework for plant phenotyping under the guidance of **Mr. Prasanjeet Betal**.

While the conceptual foundation and paper were developed by my supervisor, my contribution concentrated on implementing the **Persistent Homology pipeline, performing data preprocessing and segmentation, extracting morphological and skeleton-based features, running machine learning experiments, and analyzing results for genotype classification.**

Through this, I gained practical experience in plant image analysis, topological data representation, and ML-based phenotypic classification. The framework achieved 94% accuracy when PH features were combined with skeleton descriptors.

2. Introduction

Quantifying plant structure is critical for linking genotype to phenotype in crop science. Traditional image-based phenotyping often relies on geometric or skeleton-based descriptors, which are sensitive to noise and scale.

Under the project “**Topological Analysis of Plant Structure Using Persistent Homology**”, we aimed to integrate topological descriptors with conventional features to make genotype classification more robust and interpretable.

My work this semester primarily involved implementing the computational part of this framework — from image preprocessing to feature extraction and model training — and validating the effect of persistent homology features on classification performance.

Plant architecture determines how genetic traits translate into growth, development, and stress resilience. Accurate **morphological characterization** supports both research and crop improvement. **Morphological and skeleton-based features** offer valuable insights, especially with **deep learning** integration. However, these approaches rely on **predefined shape assumptions** and fixed spatial scales → sensitive to **noise, occlusion, and deformation**. These descriptors often overlook **the global structural integrity** of plant architecture or repeated motifs that reflect **complex growth patterns**.

Traditional geometric descriptors fail to capture the **global connectivity** and **multi-scale topological patterns** inherent in plant structures. These methods often depend on camera viewpoint, resolution, and lighting, making them less reliable for genotype-level classification. This gap motivates the use of **Topological Data Analysis (TDA)**, which studies structural features that persist across scales, independent of geometry or coordinate systems.

Persistent Homology (PH), a core concept in Topological Data Analysis, provides a scale-invariant representation of shape by tracking how connected components, loops, and voids appear and disappear as spatial resolution changes. These topological descriptors capture both local and global structure, offering a more stable and interpretable characterization of plant morphology.

The goal of this project is to combine **morphological descriptors** (e.g., skeleton-based features) with **topological descriptors** derived from Persistent Homology to build a robust, noise-resilient framework for plant genotype classification from RGB images.

In this semester, **my contribution** centered around the *implementation and experimental validation* of this framework. I developed the preprocessing pipeline for image segmentation, implemented Persistent Homology computation using Python libraries such as **GUDHI** and **Ripser.py**, extracted topological features, integrated them with skeleton-based descriptors, and trained machine learning models to evaluate classification performance.

3. My Role and Objectives

My contributions this semester can be divided into three broad categories — **conceptual understanding, implementation & experimentation, and analysis & reporting**.

A. Conceptual Understanding:

- Studied the mathematical foundation of **Persistent Homology (PH)** and its role in **Topological Data Analysis (TDA)**.
- Understood how PH differs from traditional geometric methods in capturing plant morphology.
- Explored relevant literature to connect PH-based feature extraction with phenotype–genotype mapping.

B. Implementation and Experimentation

- Implemented the PH computation pipeline in Python using libraries such as **GUDHI** and **Ripser.py**.
- Designed the **preprocessing and segmentation workflow** using **PlantCV** and **OpenCV** to generate binary plant masks.
- Performed **skeletonization** and extracted structural descriptors like curvature, convex hull area, branch points, and tip counts.
- Constructed **persistence diagrams and persistence images**, and extracted **PH summary statistics**.
- Combined morphological and PH features into a unified dataset for training machine learning models (**SVM**, **Random Forest**, and **MLP**).

C. Evaluation, Analysis, and Reporting

- Conducted comparative **performance analysis** of classifiers on skeleton-only vs. skeleton + PH features.
- Interpreted **feature importance** and **UMAP-based cluster visualizations** for genotype separability.
- Prepared **presentation slides and result visuals** for internal review and publication support.
- Documented the complete workflow, challenges, and improvements for the BTP report.

*These objectives collectively led to a fully functional plant phenotype analysis pipeline that achieved a **94% genotype classification accuracy**, demonstrating the significance of integrating topological descriptors with morphological features.*

4. Work Done This Semester

4.1 Dataset Preparation and Preprocessing

I started with a dataset containing 10 genotypes of **Brassica juncea** (Indian Mustard) plants — RE-2-2, RE-8, RE-13, RE-14, RE-35-4, CN-101849, CN-105306, CN-105308, CN-101846, and CN-112920.

Each genotype had RGB images from side and top views, captured under uniform conditions (1024×768 px).

I implemented a preprocessing pipeline using PlantCV and OpenCV to:

- Crop and align plant regions.
- Enhance contrast and normalize brightness.
- Convert RGB → LAB color space for improved segmentation.
- Apply binary thresholding on the b* channel to isolate the plant structure.

To reduce background noise, I defined a Region of Interest (ROI), removing pots and borders, ensuring a clean plant mask for downstream analysis.

4.2 Skeleton and Morphological Feature Extraction

Once the binary masks were generated, I used **PlantCV's skeletonize module** to obtain the plant's **medial axis**, representing its branching structure.

I then:

- **Pruned small spurs** from the skeleton to remove noise.
- Computed **14 morphological and skeleton-based features**, including:
 - Branch point count, tip count, curvature, segment angles
 - Convex hull area, compactness, solidity, and eccentricity
- Automated the entire process using a Python script that iterated through all samples and exported feature CSVs.

This step captured **local geometric and shape descriptors**, serving as the baseline feature set for comparison.

4.3 Persistent Homology (PH) Pipeline Implementation

This was the **core technical contribution** of my semester.

I implemented the complete Persistent Homology computation using **GUDHI** and **Ripser.py** libraries.

The workflow involved:

1. Computing the **Euclidean Distance Transform (EDT)** of each plant mask.
2. Building **2D cubical filtrations** using superlevel sets of the EDT.
3. Extracting **persistence diagrams** capturing connected components (H_0) and loops (H_1).
4. Generating **persistence images (30×30)** to represent topological information in a fixed-length format.
5. Deriving **10 PH summary statistics** such as:
 - Mean and maximum persistence
 - Total persistence, loop density
 - Persistence ratio (dominance of largest feature)

These features encode the **global structure and topological complexity** of each plant, which traditional morphology features miss.

4.4 Feature Integration and Dataset Construction

After feature extraction, I combined:

- 31 morphological/skeleton features
- 10 PH summary statistics
- 900 persistence image features

to form a **931-dimensional feature vector** per image.

All features were normalized and concatenated into a single matrix, with genotype labels for supervised learning.

This integration unified **local geometry and global topology**, making the dataset suitable for multi-class classification tasks.

4.5 Model Training and Evaluation

I used **Scikit-learn** to train and evaluate three machine learning models:

- **Random Forest (100 estimators, Gini impurity)**
- **SVM (RBF kernel, C=1.0, γ =‘scale’)**
- **MLP (2 hidden layers, ReLU, Adam optimizer)**

I applied **5-fold cross-validation** to ensure robust evaluation and measured **accuracy** and **F1-score**.

Results showed:

- Skeleton-only features achieved ~88.5% accuracy.
- Adding PH features increased accuracy to **94.2%** (Random Forest).

4.6 Result Analysis and Visualization

To interpret the results, I:

- Performed **feature importance analysis** using Random Forest’s mean Gini decrease metric.
- Used **UMAP visualization** to project the 931-dimensional space into 2D.
 - Observed distinct genotype clusters when PH features were included.
- Generated plots showing persistence diagrams, feature distributions, and classifier performance.
- Compiled visual outputs for presentation slides and paper illustrations.

These analyses confirmed that topological descriptors meaningfully separated genotypes and contributed to classification interpretability.

Overall, this semester's work resulted in a complete computational pipeline — from raw plant images to topological feature extraction and supervised classification. The Persistent Homology integration led to a measurable improvement in accuracy, establishing a proof of concept for topology-aware plant phenotyping.

5. Key Learnings

During this semester, I developed both technical expertise and conceptual understanding through hands-on implementation and analysis.

The following were my major learnings from the project:

- **Understanding of Topological Invariants**

Gained a deep conceptual understanding of how *topological features* such as connected components (H_0) and loops (H_1) serve as invariant descriptors in data analysis.

Learned how Persistent Homology captures structural persistence across scales and why it is robust against geometric distortion or noise.

- **Application of Mathematical Topology to Real Image Data**

Understood how mathematical constructs like **filtrations**, **simplicial complexes**, and **persistence diagrams** can be applied to 2D plant images to derive meaningful biological interpretations.

This was my first experience bridging pure mathematical theory with applied computer vision.

- **Hands-on Experience with Image Processing and TDA Integration**

Implemented the full image-to-feature pipeline including segmentation, skeletonization, and Persistent Homology computation.

Learned to integrate outputs from geometric (PlantCV) and topological (GUDHI, Ripser.py) analyses for unified feature learning.

- **Machine Learning Experimentation and Evaluation**

Built and evaluated models (Random Forest, SVM, and MLP) to classify genotypes.

Learned how to preprocess data, standardize feature scales, and use metrics such as accuracy and F1-score for model comparison.

- **Improved Coding Proficiency and Software Ecosystem Knowledge**

Strengthened Python programming skills and gained experience with libraries such as **PlantCV**, **OpenCV**, **Scikit-learn**, **GUDHI**, and **Ripser.py**.

Learned to write modular, reproducible code for large-scale data analysis.

- **Research Presentation and Communication Skills**

Developed the ability to convert technical work into clear visual and textual

communication.

Prepared slides, figures, and written content for internal presentations and the associated research paper.

- **Analytical Thinking and Problem-Solving**

Learned how to debug complex pipelines involving image preprocessing, feature extraction, and machine learning integration.

This improved my ability to systematically troubleshoot and optimize computational workflows.

- **Interdisciplinary Perspective**

Realized how computer science concepts like topology, geometry, and machine learning can directly contribute to biological and agricultural research.

This experience motivated me to explore more interdisciplinary applications of AI.

Overall, this project helped me evolve from understanding concepts theoretically to implementing and validating them in a real-world scientific context, reinforcing both my technical foundation and research mindset.

6. Challenges Faced

During the implementation and experimentation phase, I encountered several technical and practical challenges that required iterative debugging and optimization. Each challenge contributed to a deeper understanding of both the methodology and its limitations.

1. High Computational Cost of Persistent Homology (PH) Computation

- **Challenge:** The generation of persistence diagrams for high-resolution images was computationally expensive due to the pixel-level filtration process.
- **Action Taken:** I implemented **downsampling** of binary masks and optimized the PH computation loop using efficient NumPy vectorization.
- **Outcome:** Reduced computation time per image by nearly **60%** without losing key topological information.

2. Segmentation Sensitivity to Lighting and Color Variation

- **Challenge:** Plant segmentation accuracy was inconsistent under varying lighting and background conditions, especially for side-view images.

- **Action Taken:** Tuned the thresholding parameters dynamically and experimented with different color spaces (RGB, HSV, LAB) to isolate green foliage robustly.
- **Outcome:** Achieved more consistent plant masks and improved downstream skeleton extraction accuracy.

3. Noise in Skeletonization and the Need for Pruning

- **Challenge:** Skeletons often contained unwanted small spurs and disconnected segments, introducing noise in morphological feature computation.
- **Action Taken:** Implemented **skeleton pruning** using branch-length thresholds and morphological filters to retain only major structural components.
- **Outcome:** Significantly cleaner skeletons and more reliable feature extraction (branch count, tip count, curvature).

4. High Dimensionality of the Combined Feature Set

- **Challenge:** The integrated feature vector (931 dimensions) risked overfitting during classification, especially for models like MLP.
- **Action Taken:** Applied **feature standardization**, **L2 regularization**, and experimented with dimensionality reduction (PCA, UMAP visualization).
- **Outcome:** Improved generalization and interpretability, ensuring stable cross-validation scores across classifiers.

5. Interpreting Topological Features in a Biological Context

- **Challenge:** Translating persistence-based topological measures into biologically meaningful traits (e.g., canopy density, branching pattern) was non-trivial.
- **Action Taken:** Correlated PH statistics (loop density, persistence ratio) with visual plant structures to establish interpretability.
- **Outcome:** Enhanced understanding of how topology encodes morphological complexity relevant to genotype differences.

Overcoming these challenges strengthened my ability to design efficient pipelines, interpret multi-scale data, and adapt mathematical methods to real-world biological imaging tasks.

7. Future Work

While the current framework successfully integrates morphological and topological descriptors for genotype classification, there are several potential extensions to broaden its applicability and improve its robustness.

1. Extend Persistent Homology Analysis to Temporal Image Sequences

- The current study focuses on static RGB images. Extending PH analysis to *time-series data* can capture how topological features evolve during plant growth.
- This would allow modeling of **developmental dynamics** and **stress responses**, providing a temporal dimension to phenotype characterization.

2. Integrate Deep Learning with Topological Descriptors

- Future work can explore combining **deep neural feature embeddings** (from CNNs or Vision Transformers) with **PH-based topological features**.
- This *hybrid learning approach* can enhance feature expressiveness and reduce dependence on handcrafted descriptors, improving scalability across plant species.

3. Apply the Framework to 3D Point Cloud or Multi-View Reconstruction Data

- Using 3D imaging (e.g., LiDAR, stereo vision, or structure-from-motion) will allow **3D Persistent Homology** computation, capturing volumetric traits like canopy density and internal voids.
- This can lead to a more comprehensive analysis of plant architecture beyond 2D projections.

4. Automate Feature Selection and Dimensionality Reduction

- As the integrated feature set is high-dimensional (931 features), future work could use **automated feature selection** methods such as Recursive Feature Elimination (RFE), PCA, or manifold learning (UMAP).
- This would improve computational efficiency, model interpretability, and reduce overfitting.

5. Biological Validation and Cross-Species Generalization

- The framework can be validated across multiple plant species or under different environmental conditions to test its generalizability.

- Collaborations with agricultural research groups can help correlate topological metrics with physiological or yield-related traits.

These extensions would help transform the current prototype into a scalable and general-purpose phenotyping tool that bridges **mathematical topology, machine learning, and plant biology**.

8. Conclusion

While the theoretical foundation and research paper were primarily developed by **Mr. Prasanjeet**, my contribution this semester focused on the **practical implementation, experimentation, and performance validation** of the proposed framework.

I successfully developed the **Persistent Homology (PH) feature extraction pipeline**, integrated it with conventional morphological and skeleton-based descriptors, and trained multiple machine learning models for **genotype classification**.

The integration of PH features resulted in a measurable improvement in classification accuracy, validating the potential of **topology-aware representations** for plant phenotyping.

This hands-on experience strengthened my understanding of **computational phenotyping, topological data analysis (TDA)**, and **machine learning model design**.

Through this process, I gained exposure to both the theoretical underpinnings of topological descriptors and their practical application in biological image analysis.

Overall, this project enhanced my technical competence, problem-solving ability, and appreciation for the intersection of **computer vision, mathematics, and plant science**—laying a solid foundation for future research in data-driven plant morphology analysis.

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